Multimedia Signal Processing - 2nd module

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1 Question - 10 pts

An outdoor temperature sensor needs to communicate its readings. The admissible temperature range is $x \in [-50, +50]\,^{\circ}\mathrm{C}$.

- Plot the rate-distortion function (SNR-vs-rate) assuming that data is modeled as a continuous source without memory, uniformly distributed in $[-50, 50]$.
- Plot the operational rate-distortion function (SNR-vs-rate) assuming that data is modeled as a continuous source with memory and DPCM is used. Assume that the predictor being used is $\hat{x}_i = x_{i-1}$, In this case, the prediction error e is uniformly distributed in $[-7.5, +7.5]$.

In both cases, find the rate (expressed as an integer number of bits) and the quantization step size Δ when the system is operated at a target $SNR = 30dB$.

Consider the following data sequence

$$\{20, 22, 24, 26, 32, 34, 30\}$$

- Consider a DPCM coding scheme with fixed step size $\Delta = 5$ and the number of levels $M = 3$.
- Consider a backward-adaptive DPCM coding scheme, in which $M = 3$ and

$$\begin{aligned} \Delta &= 3, \quad \text{when } \tilde{e}_{i-1} = 0 \\ \Delta &= 5, \quad \text{otherwise} \end{aligned}$$

In both cases:

- Compute the decoded sequence obtained using such DPCM scheme.
- Compute the number of bits needed to encode the data sequence above when entropy coding is disabled.
- Verify that, for this specific example, the same mean square error distortion is obtained.

Assume that the first symbol is lossless coded.

Solution

Since $x \sim U([-50, 50])$, $\sigma_x^2 = \frac{100^2}{12} = 833.3$.
The rate-distortion function of a uniformly distributed source encoded with a uniform scalar quantizer (PCM) is

$$D(R) = \sigma_x^2 2^{-2R} \tag{1}$$

Expressed as SNR

$$SNR_{PCM}(R) = 10 \log_{10} \frac{\sigma_x^2}{\sigma_x^2 2^{-2R}} = 6.02R \tag{2}$$

Since $e \sim U([-7.5, 7.5])$, $\sigma_e^2 = \frac{15^2}{12} = 18.75$. In case of DPCM coding, the operational rate-distortion function is

$$SNR_{DPCM}(R) = 10 \log_{10} \frac{\sigma_x^2}{\sigma_e^2 2^{-2R}} = 6.02R + 16.47 \tag{3}$$

In order to attain a target distortion $SNR = 30dB$, we need to spend

$$R_x = \left\lceil \frac{30}{6.02} \right\rceil = \lceil 4.98 \rceil = 5 \text{ bits} \tag{4}$$

$$R_e = \left\lceil \frac{30 - 16.47}{6.02} \right\rceil = \lceil 2.24 \rceil = 3 \text{ bits} \tag{5}$$

Hence, the number of levels is

$$M_x = 2^{R_x} = 32 \tag{6}$$

$$M_e = 2^{R_e} = 8 \tag{7}$$

The quantization step size is

$$\Delta_x = \frac{100}{M_x} = 3.12 \tag{8}$$

$$\Delta_e = \frac{15}{M_e} = 1.875 \tag{9}$$

DPCM coding
When $M = 3$ and $\Delta = 5$, a midtread uniform quantizer with granular region $[-7.5, 7.5]$ can be implemented. That is

$$\tilde{e} = -5, \quad e \in [-7.5, -2.5)$$
$$\tilde{e} = 0, \quad e \in [-2.5, +2.5)$$
$$\tilde{e} = +5, \quad e \in [+2.5, +7.5)$$

1	1	2	3	4	5	6	7
x	20	22	24	26	32	34	30
e	-	2	4	1	7	4	-5
$\tilde{e}$	-	0	5	0	5	5	-5
$\tilde{x}$	20	20	25	25	30	35	30
Δ	5	5	5	5	5	5	5
$x - \tilde{x}$	0	2	1	1	2	1	0

In the backward adaptive case, an alternative quantizer is used when $\tilde{e}_{i-1} = 0$

$$\tilde{e} = -3, \quad e \in [-4.5, -1.5)$$
$$\tilde{e} = 0, \quad e \in [-1.5, +1.5)$$
$$\tilde{e} = +3, \quad e \in [+1.5, +4.5)$$

1	1	2	3	4	5	6	7
x	20	22	24	26	32	34	30
e	-	2	4	3	4	1	-3
$\tilde{e}$	-	0	3	5	5	0	-3
$\tilde{x}$	20	20	23	28	33	33	30
Δ	5	5	3	5	5	5	3
$x - \tilde{x}$	0	2	1	2	1	1	0

In both cases, the number of bits is $R = \lceil \log_2 3 \rceil = 2$ bits/symbol.
The MSE distortion is equal to $\frac{1}{7}\left(2 \cdot 2^2 + 3 \cdot 1^2\right)$ in both cases.

2 Question - 5 pts

Illustrate motion estimation/compensation based on block-matching.

Consider an input $N \times N$ image. Let B denote the block size and W the search range. Express the computational complexity of motion estimation C_{ME} and compensation C_{MC} in terms of number of elementary operations, as a function of N, B and W (with integer pixel accuracy).

Solution

See slide set `MMSP_3.2_Video coding`.

The number of blocks in a frame is N^2/B^2. MC requires $O(B^2)$ operations per block. Hence

$$C_{MC} = O(B^2 \frac{N^2}{B^2}) = O(N^2) \tag{10}$$

That is, the cost of motion compensation grows linearly with the number of pixels, and it does not depend on the block size.
ME requires $O((2W+1)^2 B^2)$ operations per block. Hence

$$C_{ME} = O((2W+1)^2 B^2 \frac{N^2}{B^2}) = O((2W+1)^2 N^2) \tag{11}$$

3 Questions - 5 pts (each answer can be either TRUE or FALSE)

1. Let X, Y denote two sources modeling, respectively, the horizontal and vertical components of the motion vectors in Question 2, read in raster scan.
   - [T] [F] $H(X,Y) < H(X) + (Y)$
     FALSE. X and Y are statistically independent. Hence, $H(X,Y) = H(X) + (Y)$
   - [T] [F] $H(X) >= \log_2(2W+1)$
     FALSE. $H(X) <= \log_2(2W+1)$, where equality holds when all motion vectors are equally likely.
   - [T] [F] If motion is mostly due to horizontal panning, $H(Y) < H(X)$
     TRUE. When motion is due to panning, $p_Y(y)$ is peaked around zero, leading to a lower value of $H(Y)$
   - [T] [F] X is a memoryless source
     FALSE. Motion vectors of neighboring blocks are statistically dependent.
2. Consider a perceptual audio coding system
   - [T] [F] The number of bits allocated to each frequency band is inversely proportional to SMR
     FALSE. It is directly proportional to SMR.
   - [T] [F] The perceptual model and the analysis filterbank adopt the same time-to-frequency mapping
     FALSE. The perceptual model uses a different filterbank in order to match the behaviour of the inner ear.
   - [T] [F] MDCT achieves almost perfect reconstruction
     FALSE. MDCT do achieve perfect reconstruction, provided that the TDAC condition is satisfied by the selected window.
3. Consider a block-based image coding system
   - [T] [F] With zonal quantization, each block is allocated the same bit budget
     TRUE. $R_{i,j}$ depends on $\sigma^2_{i,j}$, and it is the same for all blocks.
   - [T] [F] Decreasing the block size, the computational complexity decreases
     FALSE. The complexity is $(N^2/B^2 \cdot B^2 \log B = N^2 \log B)$
   - [T] [F] RGB to YC_bC_r mapping addresses inter-pixel correlation.
     FALSE. Colorspace mapping addresses inter-component correlation.

4 Question - 12 pts

Load data by means of the Matlab command `wavread('stereosynth.wav')`. The file contains notes played on a stereophonic synthesizer.

- Separate the left and right channels and plot them.
- Segment each channel into frames of $T = 0.1$ seconds. For each frame i, estimate the pitch $p_L(i)$ (left channel) and $p_R(i)$ (right channel) with a technique of your choice in the range [30, 300] Hz. Assume integer pitch frequency values.
- Plot the two pitch frequency sequences on a stem plot.
- Compute the conditional entropy $H(L|R)$, where L and R denote, respectively, the left and right channel pitch frequency sequences.

Solution

```
clear all;
close all;
clc;

[x fs] = wavread('stereosynth.wav');
x_r = x(:,1);
x_l = x(:,2);

subplot(211); plot(x_r);
subplot(212); plot(x_l,'r');

T = 0.10;
W = round(T*fs);

Nwin = floor(length(x(:,1))/W);
lb = round(fs/300);

p_r = NaN*ones(Nwin,1);
p_l = NaN*ones(Nwin,1);
for i = 0:Nwin-1
    x_rb = x_r(W*i+1 : W*(i+1));
    x_lb = x_l(W*i+1 : W*(i+1));

    %estimate pitch of current frames
    [r, lags] = xcorr(x_rb);
    r = r(lags>0);
    [v ind]= max(r(lb:end));
    p_r(i+1) = round(fs/(lb+ind));

    [r, lags] = xcorr(x_lb);
    r = r(lags>0);
    [v ind]= max(r(lb:end));
    p_l(i+1) = round(fs/(lb+ind));

end

figure(2)
stem(p_r);
hold on
stem(p_l,'r');

hr = hist(p_r);
p = hr./sum(hr);
p = p(p~=0);
H_r = -sum(p.*log2(p));

hl = hist(p_l);
p = hl./sum(hl);
p = p(p~=0);
H_l = -sum(p.*log2(p));

%joint entropy
hrl = hist3([p_r p_l]);
p = hrl./sum(sum(hrl));
p = p(p~=0);
H_rl = -sum(p.*log2(p));

%conditional entropy
Hl_r = H_rl - H_r;

display(['H(L) = ' num2str(H_l)]);
display(['H(R) = ' num2str(H_r)]);
display(['H(L,R) = ' num2str(H_rl)]);
display(['H(R | L) = ' num2str(Hl_r)]);
```